OpenCart E-Commerce Platform: Test Plan & Test Report

Test Plan

1. Objective

- **Project Name**: OpenCart Storefront QA Validation
- **Document Version**: 1.0
- **Prepared By**: Thulasi
- **Objective**: Define the scope, strategy, environment setup, schedule, resource allocation, and entry/exit criteria for executing end-to-end functional and regression testing across core customer journeys on the OpenCart web application.

2. Scope of Testing

In-Scope Items

1. **User Account & Authentication**
2. **Catalog & Discovery**
3. **Cart & Wishlist Order**
4. **Processing & Account History**

Out-of-Scope Items

1. **Payment gateway**
2. **Recurring automated subscription**
3. **Multi-language localization and RTL rendering tests.**
4. **Backend administrative database migration scripts.**

3. Test Strategy & Methodologies

Testing Type	Description	Objective
Functional Testing	Smoke, Sanity, Boundary, and validation checks.	Verify UI behavior matches business requirements.
End-to-End Testing	Register → Search → Add to Cart → Checkout → Order History.	Ensure uninterrupted transactional flow across all subsystems.
Regression Testing	Re-executing passing suites after fixes.	Ensure defect fixes do not destabilize existing stable modules.

4. Resources

Hardware

Resource	Quantity	Purpose
Windows 11 Systems	5	Manual testing, browser compatibility, automation execution
macOS Systems	1	Safari browser validation and iOS web testing

Software

Tool	Purpose
Google Chrome	Functional Testing
Mozilla Firefox	Cross Browser Testing
Microsoft Edge	Browser Compatibility Testing
Safari	Safari Browser Validation
Postman	API Validation
Git	Source Code Version Control
JIRA	Defect Tracking

5. Roles & Responsibilities

Role	Responsibilities
Junior QA Engineer	Design and execute test cases, log defects, perform retesting, and update execution status.
Senior QA Engineer	Prepare test plans, review test cases, validate high-priority features, and prepare execution reports.
QA Lead	Allocate work, monitor progress, coordinate with developers, and review defects.
Test Architect	Define automation standards, review testing strategy, and approve QA deliverables.
QA Manager	Monitor project quality, approve release readiness, and provide final QA sign-off.

6. Entry & Exit Criteria

Entry Criteria

- Development completed successfully.
- Unit Testing completed.
- Integration Testing completed.
- QA Build deployed.
- Test Environment configured.
- Test Data available.
- Test Cases reviewed and approved.

Exit Criteria

- 100% of planned test cases are executed.
- 0 open Blocker or Critical defects remaining.
- All Major defects either resolved or granted explicit stakeholder deferral.
- Test Summary Report finalized and accepted.

7. Schedule

Feature / Activity	Start Date	End Date	Members
Test Plan Preparation & Review	17-Jul-2026	17-Jul-2026	Thulasi / Arun Kumar
Test Scenario & Test Case Preparation	18-Jul-2026	19-Jul-2026	Thulasi
User Registration & Authentication	20-Jul-2026	21-Jul-2026	Thulasi / Kavya
Search & Product Display Page	22-Jul-2026	23-Jul-2026	Kavya
Product Compare & Wishlist	24-Jul-2026	24-Jul-2026	Kavya
Shopping Cart Operations & Boundary Tests	25-Jul-2026	26-Jul-2026	Thulasi
Checkout Flow	27-Jul-2026	28-Jul-2026	Thulasi / Rohit
My Account & Profile Updates	29-Jul-2026	29-Jul-2026	Rahul
Defect Triage, Retesting & Regression	31-Jul-2026	01-Aug-2026	All QA Members

Final Test Report, Sign-Off & GitHub Publication	01-Aug-2026	02-Aug-2026	Thulasi / Arun Kumar
Production Release	02-Aug-2026	02-Aug-2026	Release Team

8. Risks & Mitigation

Risk	Mitigation
Delay in QA Build Delivery	Revise execution schedule and prioritize smoke testing immediately after receiving the build.
Requirement Modifications	Update impacted test cases and perform regression testing before release.
Critical Functional Defects	Coordinate with the development team for immediate fixes and prioritize verification.
QA Resource Unavailability	Redistribute testing tasks among available team members to avoid schedule slippage.

Test Report

The **OpenCart Storefront** was tested across core functional modules, including user authentication, product search, wishlist, shopping cart, checkout, and order history. A total of **20 test cases** were executed, with **12 passed and 8 failed**. During testing, **8 defects** were identified: **2 Blocker, 3 Critical, 2 Major, and 1 Minor**. All identified defects were documented with their severity and current status for further resolution and retesting.

Test Execution Summary

Metric	Count
Total Test Cases	20
Executed	20
Passed	12
Failed	8

Defect Summary

Severity	Count
Blocker	2
Critical	3
Major	2
Minor	1
Total Defects	**8**

Priority Summary

Priority	Count
P0 – Blocker	2
P1 – High	3
P2 – Medium	2
P3 – Low	1

Defect Description

Defect ID	Summary	Severity	Status
DEF001	User authentication fails with 500 Internal Server Error upon clicking login button	Blocker	Open
DEF002	Search functionality crashes with unhandled on keyword search	Critical	Open
DEF003	In-Stock badge overlaps with product title on product detail page	Minor	Open
DEF004	Items clicked as 'Add to Wish List' do not persist to user Wishlist	Major	Open
DEF005	Shopping cart totals do not recalculate when line item quantity is modified	Critical	Open
DEF006	Application hangs and freezes indefinitely upon clicking 'Place Order' at checkout	Blocker	Open

DEF007	Order confirmation email is never dispatched after successful transaction	Major	Open
DEF008	My Orders / Order History page throws HTTP 500 error when viewed	Critical	Open